

Medical Technology

Medical Technology is indispensable to healthcare delivery. It is an integral part of the entire healthcare lifecycle, from the stage of screening/diagnosing to treatment/care; from restoring patients to normal lives and monitoring their health post treatment. For decades now, with advancement of technology and increasing sophistication of medical devices, the industry has played a major role in bringing down the incidence of disease and improving the overall healthcare system across the world. Globally, the healthcare sector is witnessing rapid advancements in how patients are diagnosed and treated and the Medical Technology industry is at the core of this transformation. The industry is well positioned to revolutionize healthcare in India as well. Using a mix of nascent as well as advanced technologies such as artificial intelligence, remote sensing, 3D printing, e-ICUs, virtual reality, telemedicine and others has the potential to help India's healthcare industry overcome many of the challenges faced by it- be it accessibility.

1. Affordability or quality Health care technology is capable of providing cost-effective solutions at a time when the demands on health
2. Social care services continue to increase. At the moment, the pace of change in the technology is increasing at an exponential rate.
3. Rapid technological change will require retraining and rethinking the roles of clinicians. Both existing and future generations of clinicians will need training in digital literacy to be able to make use of the 3 technology in providing optimum care. The future of healthcare lies in working hand-in-hand with technology
4. Healthcare workers have to embrace emerging healthcare technologies in order to stay relevant in the coming years

5. Taking all this into consideration, Public Health Foundation of India (PHFI), Association of Health Providers India (AHPI), Indian Institute of Science (IISc), Bangalore and Indian Institute of Space Science and Technology (IIST) have collaborated for a Certificate course on Healthcare Technology which will combine theoretical and practical aspects of healthcare technology at every level of healthcare delivery and personnel involved as per requirement.

Hookworm

- Human hookworm disease is a common helminth infection worldwide that is predominantly caused by the nematode parasites *Necator americanus* and *Ancylostoma duodenale*; organisms that play a lesser role include *Ancylostoma ceylanicum*, *Ancylostoma braziliense*, and *Ancylostoma caninum*.
- Hookworm infection is acquired through skin exposure to larvae in soil contaminated by human feces.
- Soil becomes infectious around 5-10 days after contamination and remains so for 3-4 weeks, depending on conditions

Clinical Presentation

History

- The majority of individuals who develop hookworm infection are from known endemic areas. They frequently have a history of wearing open footwear or walking barefoot in such areas.
- Most individuals with hookworm infection are asymptomatic, and diagnosis is made only by means of stool examination.
- Those symptoms that do occur depend on the type of hookworm disease present (ie, classic hookworm disease, cutaneous larva migrans,

or eosinophilic enteritis) and on the stage of the disease (ie, early or late).

Early symptoms of classic hookworm disease

- During the first 1-2 weeks after a cutaneous infection, hookworm produces a local irritation at the site of infection, termed ground itch or dew itch.
- An intensely pruritic, erythematous, or vesicular rash appears, usually on the feet or hands.
- Cough and wheezing can occur about 1 week after exposure as a consequence of larval migration through the lungs.
- Pulmonary symptoms are uncommon and usually mild, except in severe infections.
- In rare cases, severe infections may give rise to Löffler syndrome, characterized by paroxysmal attacks of cough, dyspnea, pleurisy, little or no fever, and eosinophilic pulmonary infiltrates that last several weeks after the initial infection.
- Migration of the worms into the gastrointestinal (GI) tract may cause GI discomfort secondary to irritation. As the worms mature in the jejunum, patients may experience diarrhea, vague abdominal pain, colic, flatulence, nausea, or anorexia. These symptoms are more common with initial exposures than with subsequent exposure and typically peak between 30 and 45 days after infection.
- In people who have been infected with a large burden of *A. duodenale* through oral ingestion, Wakana syndrome may occur. This syndrome resembles an immediate-type hypersensitivity reaction and is characterized by pharyngeal itching, hoarseness, nausea, vomiting, cough, dyspnea, and eosinophilia.

Later symptoms of classic hookworm disease

- Mild hookworm infections are often asymptomatic.
- Moderate-to-heavy infections cause significant blood loss, which may manifest as melena.
- Once iron reserves are exhausted, anemia develops.

- A large worm burden and a history of poor iron intake increase the likelihood of significant anemia.
- Patients with severe iron-deficiency anemia may present with lassitude, headache, palpitations, exertional dyspnea, syncope, or edema. They may also have a history of perverted taste and pica. In rare cases, anemia may provoke ischemic symptoms such as angina or claudication.
- Deficits in physical and intellectual growth can occur due to anemia and protein deficiency; these deficits may be irreversible when they develop during childhood.

Cutaneous larva migrans

Infection with zoonotic hookworms, especially *A. braziliense*, can progress with a lateral skin migration of larvae that results in the characteristic tracks of cutaneous larva migrans (creeping eruption). This is to be distinguished from the ground itch noted in classic hookworm disease.

Eosinophilic enteritis

- Eosinophilic enteritis is characterized by repeated episodes of abdominal pain in approximately 97% of affected individuals. These episodes typically occur with increasing severity and are associated with peripheral eosinophilia in almost 100% of patients and with leukocytosis in approximately 75% of patients.
- Extreme cases may mimic appendicitis or intestinal perforation.

Physical Examination

Skin and pulmonary findings are minimal. Physical findings in the early (larval migration) stage of the disease differ from those in the late (established GI infection) stage.

Early signs of classic hookworm disease

- An erythematous, pruritic, papulovesicular rash develops at the site of initial infection and may persist for 1-2 weeks after initial infection.
- Intense scratching may lead to a secondary bacterial infection, which is quite common.

- When the worms break through from the venous circulation into the pulmonary air spaces, cough, fever, and a reactive bronchoconstriction may be observed, with wheezing heard on auscultation.
- During the period of intestinal involvement, abdominal examination may reveal midepigastria pain on palpation.
- Stools may be bloody or melanotic.

Later signs of classic hookworm disease

- Signs of iron-deficiency anemia are often insensitive.
- Patients may exhibit pallor, chlorosis (greenish-yellow skin discoloration), hypothermia, spooning nails, tachycardia, or signs of high-output cardiac failure.
- Hypoproteinemia may lead to anasarca and peripheral edema.
- Poor skin texture, edema, and susceptibility to cutaneous infection suggest possible malnutrition.
- Stunted growth may be observed in children with severe infection.

Cutaneous larva migrans

- Cutaneous larva migrans manifests as pathognomonic, raised serpiginous tracts (creeping eruptions) with surrounding erythema that may last as long as 1 month if untreated.
- Lesions are most commonly seen on lower extremities but may be limited to the trunk or upper extremities, depending on the site at which the infective larvae entered the body.

Ancillary investigations –workup

Complete blood count [CBC])

- It may detect iron-deficiency anemia.
- A differential count may reveal eosinophilia (1000-4000 cells/L).

Stool for ova and parasites

- Stool should be examined for ova and parasites.
- The examination may demonstrate significant

number of hookworm eggs. However, because egg laying may be delayed, stool examination should not be considered a sensitive test for identifying hookworm infection.

- Stool examinations may have to be repeated.

PCR-based methods

- PCR-based methods for the sensitive and specific diagnosis of hookworm have been developed, but are not widely used.
- Other causes of iron loss and blood loss should be excluded.

Other investigations

- In cases of severe infection, chest radiography may show diffuse alveolar infiltrates during the migration of the larvae through the lung to the intestine.
- Once infection is established in the gut, imaging studies are not helpful.
- In cases of eosinophilic enteritis, colonoscopy may reveal ileal and colonic ulceration and, occasionally, adult hookworms.

Ascariasis/Roundworms

- Ascariasis is the most common helminthic infection, with an estimated worldwide prevalence of 804 million cases in 2013.
- It is caused by *Ascaris lumbricoides*, which is the largest parasite from the class of nematodes commonly known as roundworms.
- Usually *A. lumbricoides* or *A. suum* infection is asymptomatic.
- Ascariasis is most prevalent in children of tropical and developing countries, where they are perpetuated by contamination of soil by human feces or use of untreated feces as fertilizer and the water or food contaminated by the eggs is ingested.
- Symptomatic ascariasis occurs during the adult worm's intestinal or larval migration phase and may manifest as growth retardation, pneumonitis, intestinal obstruction, or hepatobiliary and pancreatic injury.

- Ascariasis may exist as a zoonotic infection associated with pigs and use of hog manure and is caused by *Ascaris suum*, which is very similar to *Ascaris lumbricoides* but, in most endemic areas, it is most likely transmitted from person to person.

Clinical presentation

History

- Early symptoms of ascariasis, during the initial lung migration (first 10-14 days after ingestion) include cough, dyspnea, wheezing, urticaria, hemoptysis, and chest pain.
- This may be seasonal after rains in some countries, such as Saudi Arabia. Returning travelers could also present with Loeffler syndrome of eosinophilic pneumonia if exposed in a high-risk environment.
- Abdominal pain or discomfort, distension, colic, nausea, anorexia, and intermittent diarrhea may be manifestations of partial or complete intestinal obstruction by adult worms during the late phase of infection six to eight weeks after egg ingestion.
- Adult worms are passed in the stool and can be seen on stool examination.
- Jaundice, nausea, vomiting, fever, and severe or radiating abdominal pain may suggest cholangitis, pancreatitis, hepatic abscess, or appendicitis.

Physical

- Within 1 to 2 weeks of new infection following egg ingestion, *Ascaris* larvae migrate through lungs causing eosinophilic pneumonitis. This is known as the Loeffler syndrome.
- Parasitic infections with *Strongyloides stercoralis*, *Ancylostoma duodenale*, *Necator americanus*, *Toxocara*, schistosomiasis, and lymphatic filariasis can present with a similar pathological presentation.
- Patients may present with fever, dry cough, dyspnea, wheezing, and urticaria that subsides within 5 to 10 days.
- Rales, wheezes, and tachypnea may develop

during pulmonary migration, particularly in persons with a high worm burden.

- Urticaria and fever may also develop late in the migratory phase.
- Abdominal distension is nonspecific, but is common in children with ascariasis.
- Abdominal tenderness, especially in the right upper quadrant, hypogastrium, or right lower quadrant, may suggest complications of ascariasis.
- Evidence for nutritional deficiency due to ascariasis is strongest for vitamins A and C, as well as for protein, as indicated by albumin and growth studies in children observed prospectively.

Ancillary investigations—workup

Stool examination for ova and parasites:

- Visualization of large, brown 60 μ m X 50 μ m trilayered eggs is confirmatory.
- However, stool examination may be negative for ova for up to 40 days after infection because of the time needed for migration and maturation of the worm.
- However, it may take 40 days for eggs to appear in the stools and if only infested with male worms, there may be no eggs at all.

Sputum examination: *Ascaris* larvae may be observed in microscopic wet preparations of sputum during the pulmonary migration phase.

CBC with differential: Results will reveal peripheral eosinophilia during the tissue migration phase of the infection. It may take weeks to resolve, but can be masked with the administration of steroids. It is more likely in the early lung phase than the late phase.

Serological tests: Serum IgG and IgE are elevated during early infection, but this finding is not clinically useful.

Pinworm (Enterobiasis)/ Threadworms

- Enterobius (syn. Oxyuris) vermicularis is a human-pathogenic intestinal parasite belonging to the nematodes (Nematoda).
- Synonyms include "threadworm" and "seatworm."

- Symptomatic pinworm infection is referred to as enterobiasis (older term: oxyuriasis).
- Threadworms, also called pinworms, are tiny parasites that grow to about 1 cm in length and look like tiny pieces of white thread. They live in the intestines.
- In humans, they reside in the cecum, appendix, and ascending colon.
- Female pinworms are 8-13 mm long, and males are 2-5 mm long.
- Pinworm infection is primarily a pediatric condition, and parents are typically infected via transmission through their children.
- Pinworm is prevalent throughout the temperate regions of the world and is the most common helminthic infection in the United States.
- Transmission can occur via direct contact with contaminated furniture, bedclothes, bedding, towels, toilets, doorknobs, or other objects. The parasite also can be transmitted during sexual contact.
- Pinworm infection generally is asymptomatic; asymptomatic carriers are common.
- The cure rate with treatment is 90-95%. Re-infection is common, especially if all contacts are not treated simultaneously.

Clinical presentation and diagnosis

History and Physical Examination

- The patient history of pinworm infection, also called enterobiasis, includes itching or prickling pain in the anal area (usually at nighttime or in the early morning), intense anal itching, restless sleep or difficulty sleeping, and, rarely, abdominal discomfort or loss of appetite.
- Other symptoms include irritability and, in females, vaginal itching. Most patients, however, are asymptomatic.
- The pale-colored female pinworm (10 mm) may be seen in the perianal region with the naked eye. The worm's appearance is easily confused with bits of cotton thread.
- Eggs (30 μ m X 50-60 μ m) usually are not seen without a microscope.

- Perianal excoriations secondary to scratching may be found.

Ancillary investigations and workup

Stool for ova and parasites

- Glass slide microscopic analysis for *Enterobius vermicularis* may be performed looking for ova and female pinworms.
- A specimen is best obtained by dabbing the stretched, unwashed perianal folds in the early morning with cellophane tape and affixing the specimen onto a slide. A negative test for 5 consecutive mornings effectively rules out the diagnosis.

Note-

- Stool microscopy is not a helpful diagnostic tool, since the worm eggs are deposited outside the intestine.
- Serological methods are of no diagnostic relevance.
- Similarly, neither blood eosinophilia nor elevated immunoglobulin E levels are generally expected due to the low invasiveness of the worms.

MCQs

1. Diagnosis of *Enterobius vermicularis* (pinworm) is done by
 - a. Perianal skin
 - b. Feces
 - c. Sputum
 - d. Duodenal fluid
2. Which of the following is the most common symptom of pinworm (*Enterobius vermicularis*) infection?
 - a. Ulcerative lesions in the small or large bowel
 - b. Abdominal distention and cramps
 - c. Nocturnal perianal pruritus
 - d. Vomiting and watery diarrhea
3. The causative organism for roundworm infestation
 - a. *Ascaris lumbricoides*
 - b. *Enterobius vermicularis*
 - c. *Ancylostoma duodenale*
 - d. *Necator americanus*

4. Loeffler syndrome seen in

- a. Aspergillosis
- b. Coccidioidomycosis
- c. Ascariasis
- d. Schistosomiasis

5. The stool examination for Ascariasis is positive after how many days

- a. More than 1 week
- b. More than 40 days

c. More than 14 days

d. More than 20 days

6. Wakana syndromeseen in

- a. Oral ingestion of *Ancylostomaduodenale*
- b. Cutaneous infection of *Ancylostoma-duodenale*
- c. Oral ingestion of *Enterobiusvermicularis*
- d. Cutaneous infection of *Enterobiusvermicularis*

Answer

1.	a	2.	c	3.	a
4.	c	5.	b	6.	a